

The Far End of a Schur Bridge Does Not Care How Its Clusters Are Priced, and the Interior Does

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Abstract

A two-tier allocation holds a portfolio inside each cluster and then decides how much capital each cluster gets. The second decision needs a $k \times k$ matrix, and every construction in this family so far builds that matrix from the same cluster portfolios it holds. It need not. Separating what is held from what it is priced against gives a family with two spaces rather than one, and the far end of the bridge is exact for every choice of the pricing space: the k conditioned directions already sum to $\Sigma^{-1}u$, so any non-degenerate pairing recovers it. Choosing one representative asset per cluster, as hierarchical core-orbital allocation does, therefore cannot affect the far end. The interior is a different matter. Once the two spaces differ the $k \times k$ matrix is no longer symmetric, and it can pass through singularity inside the dial: we give a positive-definite 6×6 example where it does, and on estimated covariances it does so in about a quarter of markets. Pricing against the cluster portfolios keeps the matrix a Gram matrix of Σ and removes the possibility.

1 Introduction

A two-tier allocation makes two decisions. Inside cluster C it holds some portfolio, and across clusters it decides how much capital each one gets. The second decision is an allocation problem in k dimensions, and it needs a $k \times k$ matrix to work with.

Every construction in this family builds that matrix from the cluster portfolios themselves. Nested clustered optimization [6] uses the covariance of the cluster portfolios; hierarchical risk parity [5] and hierarchical equal risk contribution [7] use variances computed from the same holdings. Written in the pair form of [2], all of them set the pricing object equal to the holding.

Hierarchical core-orbital allocation [1] does not. It selects one physical asset per cluster, allocates across those k assets, and only then redistributes each cluster's capital among its members. What is priced and what is held are different objects, and the reason given is institutional: the strategic decision is expressed in instruments a desk can trade.

This note takes that separation seriously and asks what it costs. The answer is nothing at the far end and something in the interior. Section 3 shows that the far end is exact for every pricing space, because the k conditioned directions already sum to the global optimum. Section 4 shows that once the two spaces differ the $k \times k$ matrix loses its symmetry and can become singular inside the dial. Section 5 reads core-orbital allocation in these terms.

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2 Two spaces

Let $\Sigma \succ 0$ be the covariance of n assets, let $u \in \mathbb{R}^n$ be a companion vector, and let $C_1, \dots, C_k$ partition the assets. For a cluster C write $-C$ for its complement and, for $\gamma \in [0, 1]$, define the conditioned pair

$$Q_C(\gamma) = \Sigma_{CC} - \gamma \Sigma_{C,-C} \Sigma_{-C,-C}^{-1} \Sigma_{-C,C}, \quad b_C(\gamma) = u_C - \gamma \Sigma_{C,-C} \Sigma_{-C,-C}^{-1} u_{-C}, \quad (1)$$

and the direction $d_C(\gamma) = Q_C(\gamma)^{-1} b_C(\gamma)$, embedded in $\mathbb{R}^n$ with support C . Collect these as the columns of $D_\gamma \in \mathbb{R}^{n \times k}$.

Definition 1 (Trial and test spaces). *The trial matrix is D_γ : its column span is what the portfolio is built from. The test matrix is any $R \in \mathbb{R}^{n \times k}$: it is what the trial directions are priced against. The allocation is*

$$w(\gamma, R) \propto D_\gamma (R^\top \Sigma D_\gamma)^{-1} R^\top u, \quad (2)$$

normalized to sum to one, and defined when $R^\top \Sigma D_\gamma$ is invertible.

Taking $R = D_\gamma$ recovers the constructions already in the literature. The $k \times k$ matrix is then $D_\gamma^\top \Sigma D_\gamma$, and (2) is the minimum-variance portfolio on the span of the trial directions. At $\gamma = 0$ with $u = \mathbf{1}$ that is nested clustered optimization. We call this the *matched* case, and any $R \neq D_\gamma$ an *unmatched* one.

3 The far end

Proposition 1 (The far end is free of the test space). *Suppose $R^\top \Sigma D_1$ is invertible. Then $w(1, R) \propto \Sigma^{-1} u$, whatever R is.*

Proof. Block inversion on the partition $(C, -C)$ gives $(\Sigma^{-1} u)_C = Q_C(1)^{-1} b_C(1) = d_C(1)$ [2, App. A]. The supports $C_1, \dots, C_k$ partition the index set, so summing the columns of D_1 reconstructs the whole vector: $D_1 \mathbf{1}_k = \Sigma^{-1} u$. Multiplying on the left by $R^\top \Sigma$,

$$R^\top \Sigma D_1 \mathbf{1}_k = R^\top \Sigma \Sigma^{-1} u = R^\top u,$$

so $(R^\top \Sigma D_1)^{-1} R^\top u = \mathbf{1}_k$ and the right side of (2) is $D_1 \mathbf{1}_k = \Sigma^{-1} u$. $\square$

The proposition says something slightly surprising about where the work is done. At $\gamma = 1$ the trial directions are not merely a subspace containing the answer, they are its blocks, and the coefficients that assemble them are all one. Any invertible pairing recovers those coefficients, so the pricing space carries no information at the far end at all.

Corollary 1 (Representative selection does not reach the far end). *Let $E \in \mathbb{R}^{n \times k}$ have $E_{p_c, c} = 1$ for one chosen asset p_c per cluster and zeros elsewhere. Then $w(1, E) \propto \Sigma^{-1} u$ for every selection $p_1, \dots, p_k$ for which $E^\top \Sigma D_1$ is invertible.*

A rule that chooses the representative, whether by medoid, by alignment with the cluster's first principal component, or by maximal squared correlation with the rest of the cluster [1], therefore decides where the interior of the bridge runs and nothing about its far end. The certificate checks proposition 1 for five test spaces, including two different representative selections and a dense random one, and for a general companion vector as well as $u = \mathbf{1}$.

Remark 1 (Which far ends). *With $u = \mathbf{1}$ the far end is minimum variance, with $u = \sigma$ maximum diversification, and with $u = \mu$ the tangency direction; the proof uses no property of u . Proposition 1 needs the conditioning of (1) to be on every other asset. Conditioning instead on one knot per other cluster is exact under the model of [3], and proposition 1 then holds under that model.*

4 The interior

The freedom of proposition 1 does not extend inward, and the reason is structural rather than statistical.

Proposition 2 (The matched case has no pole). *If D_γ has full column rank then $D_\gamma^\top \Sigma D_\gamma$ is symmetric positive definite. Hence (2) with $R = D_\gamma$ is defined at every γ for which the pairs (1) exist.*

Proof. It is a Gram matrix: $x^\top D_\gamma^\top \Sigma D_\gamma x = (D_\gamma x)^\top \Sigma (D_\gamma x) > 0$ for $x \neq 0$ since $\Sigma \succ 0$ and $D_\gamma x \neq 0$. $\square$

When $R \neq D_\gamma$ the matrix $R^\top \Sigma D_\gamma$ is not symmetric and proposition 2 does not apply. Its determinant is a rational function of γ and may vanish in $[0, 1]$ on ordinary positive-definite input. Take

$$\Sigma = \begin{pmatrix} 20 & 8 & -5 & 5 & 5 & -6 \\ 8 & 36 & -11 & 8 & 7 & -12 \\ -5 & -11 & 33 & 2 & 6 & 16 \\ 5 & 8 & 2 & 16 & 3 & 9 \\ 5 & 7 & 6 & 3 & 29 & -18 \\ -6 & -12 & 16 & 9 & -18 & 50 \end{pmatrix},$$

with clusters $\{1, 2, 3\}$ and $\{4, 5, 6\}$, representatives 3 and 4, and $u = \mathbf{1}$. Its leading principal minors are positive. The determinant of $E^\top \Sigma D_\gamma$ is positive at $\gamma = \frac{1}{2}$ and negative at $\gamma = \frac{3}{5}$, so it vanishes between them and the allocation has a pole strictly inside the dial. On the same input the matched determinant stays positive throughout. Both facts are certified in exact arithmetic.

How often this bites. A pole is not a curiosity of a constructed example. Take twenty-four assets in four clusters of six, drawn from a one-factor-per-cluster model, estimate the covariance from one hundred and twenty observations, and scan the dial on a grid of four hundred points. The determinant of the outer matrix changes sign somewhere in $[0, 1]$ in 28% of markets when the test space is one asset per cluster and in 35% when it is dense random. In the matched case it never does, as proposition 2 requires.

What it costs out of sample. The same markets, scored by the true variance of the estimated portfolio in excess of the true optimum, give the median profile in table 1. At the far end all four agree, exactly as proposition 1 says, and they agree on a number well above zero because the covariance is estimated. Away from it the unmatched choices are not merely worse on average, they are occasionally enormous, and the excursions sit where the determinant crosses.

test space	$\gamma = 0$	$\gamma = 0.2$	$\gamma = 0.6$	$\gamma = 1$
cluster portfolios (matched)	10.5	9.9	12.2	23.4
one asset per cluster, maximal squared correlation	37.3	57.0	112.9	23.4
one asset per cluster, chosen at random	21.8	20.5	289.3	23.4
dense random	751.4	265.5	1112.0	23.4

Table 1: Excess variance over the true optimum, in percent, mean over thirty estimated markets. The far-end column is identical across rows by proposition 1.

So the separation of the two spaces buys exactness that is free and stability that is not. A test space chosen for interpretability is safe at the far end and exposed everywhere else.

5 Core-orbital allocation

Hierarchical core-orbital allocation [1] selects a core asset p_c in each cluster, allocates capital W across the k cores using their own covariance $\Sigma_{\mathcal{R}\mathcal{R}} = E^\top \Sigma E$, and then redistributes each W_c among the members of cluster c in proportion to a portfolio computed from that cluster's own block. Writing V for the matrix of those normalized cluster portfolios, the result is $w = VW$.

In the notation of (2) the trial matrix is D_0 , whose columns are the cluster portfolios up to scale, and the test matrix is E . The outer matrix, however, is $E^\top \Sigma E$ and not the pairing $E^\top \Sigma D_0$. The method is therefore adjacent to the family rather than inside it, and the difference is one factor, which the rest of this section makes precise.

Proposition 3 (When the two coincide). *Let $N = \text{diag}(\mathbf{1}^\top d_{C_c}(0))$, so that $D_0 = VN$. Then $E^\top \Sigma D_0 = (E^\top \Sigma V)N$, and core-orbital allocation agrees with the member $w(0, E)$ of (2) for every companion vector if and only if*

$$E^\top \Sigma (V - E) = 0, \quad (3)$$

that is, if and only if each cluster's holding has the same covariance with every core as that core has.

Condition (3) holds trivially when every cluster is a singleton, where the holding is its own core and core-orbital allocation is the far end itself. Under a one-factor model within each cluster it says that each holding carries the same factor exposure as its core. It does not hold in general, and the certificate exhibits a case where it fails.

Which matrix to price with. Pricing with $E^\top \Sigma E$ rather than with $E^\top \Sigma D_0$ places the method outside the family, so proposition 1 does not apply to it. That is not a defect to be repaired. A method fixed at $\gamma = 0$ never visits the far end, so the exactness it forgoes is of no use to it, while $E^\top \Sigma E$ is a principal submatrix of Σ and therefore symmetric positive definite, which is exactly what section 4 shows the pairing is not. Substituting the pairing at $\gamma = 0$ on estimated covariances lowers the median excess variance and raises the worst case by an order of magnitude in our runs, so the substitution is not one we would recommend.

What the choice does cost. Under a one-factor model within each cluster the covariance of the holdings satisfies $V^\top \Sigma V = K \Sigma_{\mathcal{R}\mathcal{R}} K + \Delta$, with K diagonal in each cluster's factor exposure and Δ diagonal, so pricing by the cores omits both. Minimum variance is invariant to a common scaling, so the omission is immaterial when the clusters carry similar exposures and grows with their dispersion. On simulated universes whose clusters resemble one another, as the maturity and credit segments of a bond curve do, the two price vectors differ by half a percentage point of weight and their out-of-sample variances by a fraction of a percent; on universes whose clusters differ sharply in exposure and volatility the gap is several times larger. The comparability with nested clustered optimization reported in [1] on a fixed-income universe is what this predicts, and an equity universe with dispersed cluster exposures is where the prediction could fail.

6 Closing

The bridges catalogued so far are named by their two ends, and each assumes that a cluster is priced by the portfolio held in it. Dropping that assumption adds a second axis rather than another row. Along it the far end is not a point but a face: any non-degenerate pricing space reaches $\Sigma^{-1}u$, so the choice of representative is a statement about the interior alone.

Two questions are left open. The first is whether any pricing space improves on the matched one in the interior, where the evidence here is that none of the three tried does. The second is whether a pricing space can be chosen to avoid the pole while keeping exactness, for instance by requiring $R^\top \Sigma D_\gamma$ to be positive definite along the path; the matched choice satisfies this by construction, and whether any other does is not settled here.

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